

An Empirical Comparison of Knowledge-Distillation Objectives for Compressing a Pretrained Language Model

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Abstract

Knowledge distillation (KD) transfers the behavior of a large “teacher” language model into a smaller “student.” The literature offers many objectives—forward and reverse Kullback–Leibler divergence on output logits, symmetric divergences such as the generalized Jensen–Shannon divergence and total variation distance, intermediate-representation matching, and sequence-level distillation on teacher-generated text—but few controlled comparisons at small scale. We distill the pretrained GPT-2 (124M parameters) into a 29M-parameter, 4-layer student under an equal step budget for seven objectives, evaluating WikiText-2 perplexity, LAMBADA final-word accuracy, and HellaSwag and ARC-Easy multiple-choice accuracy. We additionally investigate an auxiliary representation-alignment target motivated by a user-supplied reference to an Anthropic “J-Space” paper; we document that no such paper exists in the public record and instead evaluate the closest defensible interpretation. **[RESULTS SUMMARY TO BE FILLED FROM runs/summary.json]**

1 Introduction

Large language models are expensive to serve; distillation [1] compresses them by training a small student to imitate a large teacher rather than only the raw data. Since Hinton et al.’s original formulation, the field has produced a family of objectives that differ chiefly in (i) *what* is matched (output distributions, intermediate states, or generated sequences) and (ii) *which divergence* measures the mismatch. This paper implements the most common objectives in a single controlled harness and compares them when distilling a pretrained GPT-2 teacher on a fixed compute budget.

2 Methods

Let $p(\cdot | x)$ be the teacher’s next-token distribution and $q_\theta(\cdot | x)$ the student’s. All students share one architecture, initialization, optimizer, and step budget; only the loss differs.

Hard-label baseline (hard). Standard next-token cross-entropy on the data, $\mathcal{L}_{\text{CE}} = -\log q_\theta(y_t | x_{<t})$, with no teacher involvement.

*Experiments, implementation, and drafting assisted by Claude (Anthropic).

Forward-KL logit distillation (kd). The classic objective [1]: with temperature T and mixing weight α ,

$$\mathcal{L} = \alpha \mathcal{L}_{\text{CE}} + (1 - \alpha) T^2 \text{KL}\left(p^{1/T} \parallel q_\theta^{1/T}\right), \quad (1)$$

where $p^{1/T}$ denotes the temperature-flattened distribution. Forward KL is *mass-covering*: the student is penalized wherever the teacher assigns probability the student misses.

Reverse-KL distillation (rkl). $\text{KL}(q_\theta \parallel p)$ is *mode-seeking* and was argued by MiniLLM [5] to suit generative students that cannot cover every teacher mode. We use the teacher-forced variant (computed on data prefixes rather than on-policy student rollouts) to keep the comparison budget-matched.

Generalized Jensen–Shannon divergence (jsd). Following GKD [6], $\text{JSD}_\beta(p, q) = \beta \text{KL}(p \parallel m) + (1 - \beta) \text{KL}(q \parallel m)$ with $m = \beta p + (1 - \beta)q$; $\beta = 0.5$ gives the symmetric JSD, and the family interpolates between forward ($\beta \rightarrow 1$) and reverse ($\beta \rightarrow 0$) KL.

Total variation distance (tvd). $\frac{1}{2} \sum_v |p_v - q_v|$, a bounded symmetric alternative studied in f-divergence distillation [7].

Hidden-state matching (hidden). TinyBERT/DistilBERT-style [4, 3] intermediate-layer transfer: each student layer s is mapped to teacher layer $t(s)$ (uniform spacing), a learned linear projection lifts the student width (384) to the teacher width (768), and a scale-normalized MSE between projected student and teacher states is added to the forward-KL objective.

Sequence-level KD (seqkd). Kim & Rush [2]: the teacher generates full sequences (here, sampled completions of 16-token WikiText-2 prompts, top- $k=50$), and the student trains with plain cross-entropy on this synthetic corpus.

J-space auxiliary target (jspace). [TO BE WRITTEN AFTER RESEARCH: what was found about the claimed Anthropic paper, the chosen interpretation, and its formula.]

3 Experimental Setup

Teacher. GPT-2 124M (12 layers, 768-dim), pretrained weights in HF format.¹ **Student.** 4 layers, 384-dim, 6 heads, GPT-2 BPE vocabulary (50,257), $\sim 29\text{M}$ parameters ($\sim 23\%$ of teacher), randomly initialized, identical seed across methods. **Data.** WikiText-2 (word-level release), packed into 128-token blocks; 4,000 training blocks. **Optimization.** AdamW, lr 3×10^{-4} with linear warmup (5%) and linear decay, weight decay 0.01, gradient clipping 1.0, batch size 8, 600 steps ($\approx 0.6\text{M}$ training tokens), $T=2$, $\alpha=0.5$. Single seed (compute-constrained; CPU-only). **Evaluation.** WikiText-2 test perplexity on 200 held-out blocks; LAMBADA final-word greedy accuracy (300 examples); HellaSwag and ARC-Easy accuracy via length-normalized log-likelihood over answer choices (300 examples each).

¹Weights obtained from the legacy Hugging Face S3 mirror; the experiment environment blocks huggingface.co.

4 Results

[TABLE + ANALYSIS TO BE FILLED FROM runs/summary.json]

5 Discussion and Limitations

[TO BE WRITTEN: interpretation, capacity-gap notes, small-budget caveats, single-seed caveat, teacher-forced rkl caveat.]

References

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- [5] Y. Gu, L. Dong, F. Wei, M. Huang. MiniLLM: Knowledge distillation of large language models. *ICLR*, 2024.
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